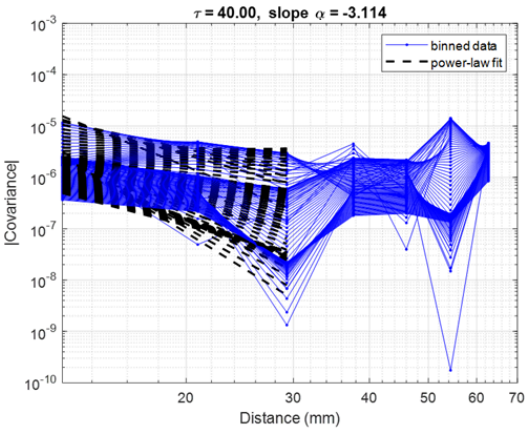
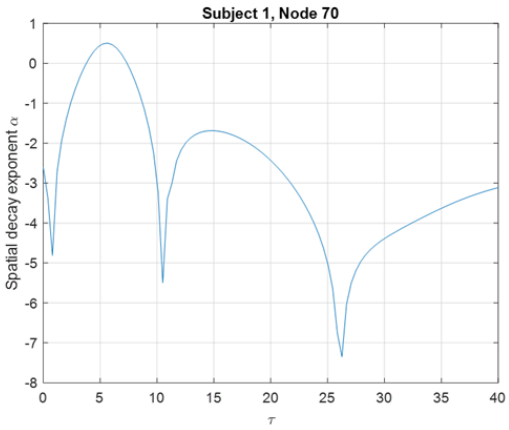
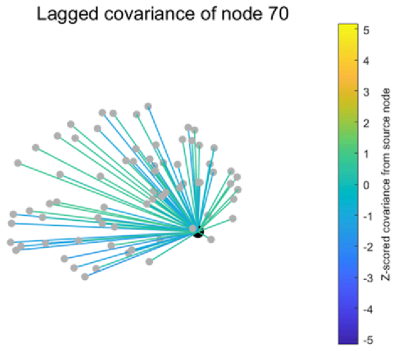
log2

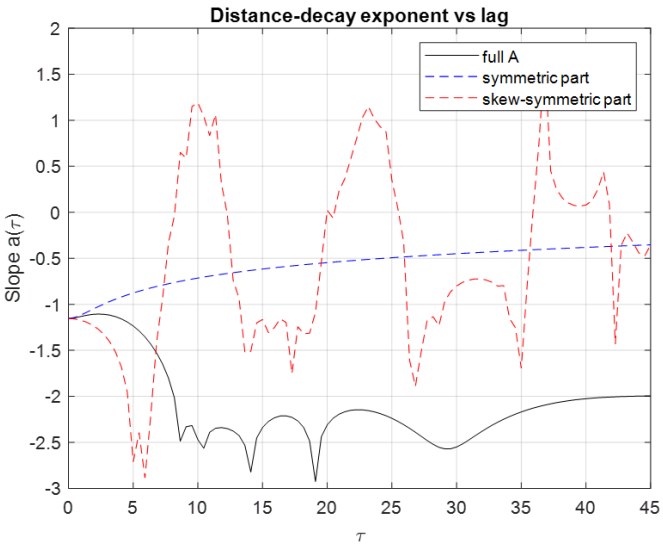




for single node the averaged covariance (concentric propagation assumption) over binned distance is more evident. HERE ABSOLUTE COVARIANCE (correlation and not type of correlation)



Interesting to determine whether the concentric propagation assumption holds or not, and in fact if we plot between single nodes, we see that there are different orientations for the covariance for a specific lag. Averaging out causes the loss of information about propagation shape? HERE SIGNED COVARIANCE



Now it would make sense to see the same evolution of $P_{ij}(\tau)$ with respect to the different components of its dynamics A. Given

$$A = \left(-\frac{1}{2}\Sigma_w + S\right)\Sigma^{-1} = \underbrace{-\frac{1}{2}\Sigma_w\Sigma^{-1}}_{A_{\text{sym}}} + \underbrace{S\Sigma^{-1}}_{A_{\text{skew}}}$$

I can decompose the dynamics of the covariance matrix into its dissipative and rotational components:

- **Symmetric Evolution**: Drives pure diffusion or decay of correlations. Its propagator will tend monotonically toward zero off-diagonals without any oscillatory or rotational behavior.

$$P_{\text{sym}}(\tau) = \exp(A_{\text{sym}}\tau)\Sigma = \exp\left(-\frac{1}{2}\Sigma_w\Sigma^{-1}\tau\right)\Sigma$$

- **Skew-Symmetric Evolution**: Drives pure rotation or phase-mixing in the Σ-metric. Its propagator rearranges patterns without decay.

$$P_{\text{skew}}(\tau) = \exp(A_{\text{skew}}\tau)\Sigma = \exp(S\Sigma^{-1}\tau)\Sigma$$

Generally, linearity of the evolutions doesn't hold $P(\tau) \neq P_{\text{sym}} + P_{\text{skew}}$. By measuring the distance-decay exponent $\alpha(\tau)$ for both P_{sym} and P_{skew} you see how much of your empirical decay is due to diffusion vs to the underlying mixing or oscillatory turnover. Often the skew part by itself shows almost no long-range decay, but there can be lags where it actually sharpens or spreads patterns transiently.